

Lab Manual — Data Types in Python

Module 1 | Python Programming Fundamentals

Topic 2 of 3: Data Types

Background

Every value in Python has a **data type** — it tells Python what kind of data you're working with and what operations are allowed on it.

Core Built-in Types

Type	Example	Description
int	42, -7	Whole numbers
float	3.14, -0.5	Decimal numbers
str	"hello"	Text / characters
bool	True, False	Logical values
list	[1, 2, 3]	Ordered, mutable collection
tuple	(1, 2, 3)	Ordered, immutable collection
dict	{"key": "value"}	Key-value pairs
NoneType	None	Absence of a value

Checking Types

Use the built-in `type()` function:

```
print(type(42))      # <class 'int'>
print(type(3.14))    # <class 'float'>
print(type("hello")) # <class 'str'>
```

Type Conversion (Casting)

You can convert between types:

```
int("5")      # → 5
float(3)       # → 3.0
str(100)       # → "100"
bool(0)        # → False
bool(1)        # → True
```

Solved Tasks

Task 1 — Identifying Types

Objective: Use `type()` to inspect several values.

```
values = [42, 3.14, "Python", True, None, [1, 2], (3, 4), {"a": 1}]

for v in values:
    print(f"Value: {v!r:20} | Type: {type(v).__name__}")
```

Output:

```
Value: 42 | Type: int
Value: 3.14 | Type: float
Value: 'Python' | Type: str
Value: True | Type: bool
Value: None | Type: NoneType
Value: [1, 2] | Type: list
Value: (3, 4) | Type: tuple
Value: {'a': 1} | Type: dict
```

Task 2 — Type Conversion

Objective: Convert values between types and handle a failed conversion.

```
# int → float → str
num = 7
print(type(num), num)

num_float = float(num)
print(type(num_float), num_float)

num_str = str(num_float)
print(type(num_str), num_str)

# String to int
age_input = "22"
age = int(age_input)
print("Next year:", age + 1)
```

Output:

```
<class 'int'> 7
<class 'float'> 7.0
<class 'str'> 7.0
Next year: 23
```

Task 3 — Strings and Their Properties

Objective: Explore common string operations.

```
name = "Artificial Intelligence"

print(len(name))          # Length
print(name.upper())       # Uppercase
print(name.lower())       # Lowercase
print(name.replace("Intelligence", "Neural Networks"))
print(name[0:10])         # Slicing
print("AI" in name)       # Membership test
```

Output:

```
23
ARTIFICIAL INTELLIGENCE
artificial intelligence
Artificial Neural Networks
Artificial
False
```

Task 4 — Booleans and Truthiness

Objective: Understand which values Python considers True or False.

```
test_values = [0, 1, -5, "", "hello", [], [1, 2], None]

for val in test_values:
    print(f"bool({val!r}) = {bool(val)}")
```

Output:

```
bool(0) = False
bool(1) = True
bool(-5) = True
bool('') = False
bool('hello') = True
bool([]) = False
bool([1, 2]) = True
bool(None) = False
```

Task 5 — Working with None

Objective: Use None as a placeholder and check for it correctly.

```
result = None

if result is None:
    print("No result yet – computation pending.")
else:
    print("Result:", result)

# Assign and check again
result = 42
if result is None:
    print("Still nothing.")
else:
    print("Got a result:", result)
```

Output:

```
No result yet – computation pending.
Got a result: 42
```

Student Practice Tasks

Solve the following tasks on your own. No solutions are provided here — attempt each one independently.

1. Create one variable of each core type (`int`, `float`, `str`, `bool`, `list`, `tuple`, `dict`, `NoneType`) and print both the value and its type using `type()`.
2. Ask the user to enter their age as input (which is always a string). Convert it to an integer, add 5, and print the result.
3. Create a string with your full name. Print its length, convert it to uppercase, replace your first name with "Student", and check whether the letter "a" appears in it.
4. Write a program that demonstrates the difference between `int` and `float` division: divide 7 by 2 using `/` and `//` and print both results with their types.
5. Given the string "100", convert it to an integer and a float. Print all three versions (original string, `int`, `float`) along with their types.
6. Create a list of 5 numbers. Print the list, its type, its length, and the type of the first element.
7. Create a dictionary storing a student's `name`, `age`, `grade`, and `is_enrolled` (`bool`). Print each key-value pair along with the type of the value.
8. Write a program that tests truthiness: for each of the following — `0`, `0.0`, `""`, `"0"`, `[]`, `{}`, `None`, `False` — print whether Python treats it as `truthy` or `falsy`.
9. Convert the float `9.99` to an integer and explain in a comment what happens to the decimal part. Then convert it to a string and demonstrate that you can no longer do arithmetic on it.
10. Write a program that uses `isinstance()` to check whether a given variable is of type `int`, `float`, or `str`, and prints a different message for each case.